Problem A. Who Ate the Cake?

Time Limit 2000 ms
Mem Limit 1048576 kB

Problem Statement

Takahashi's cake has been eaten by someone. There are three suspects: person 1, person 2 , and person 3.

There are two witnesses, Ringo and Snuke. Ringo remembers that person A is not the culprit, and Snuke remembers that person B is not the culprit.

Determine if the culprit can be uniquely identified based on the memories of the two witnesses. If the culprit can be identified, print the person's number.

Constraints

- $1 \leq A, B \leq 3$
- All input values are integers.

Input

The input is given from Standard Input in the following format:

A B

Output

If the culprit can be uniquely identified based on the memories of the two witnesses, print the person's number; otherwise, print `-1`.

Sample 1

Input	Output
1 2	3

From the memories of the two witnesses, it can be determined that person 3 is the culprit.

Sample 2

Input	Output
1 1	-1

From the memories of the two witnesses, it cannot be determined whether person 2 or person 3 is the culprit. Therefore, print `-1`.

Sample 3

Input	Output
3 1	2

Problem B. Forbidden List

Time Limit 2000 ms
Mem Limit 1048576 kB

Problem Statement

Given are an integer X and an integer sequence of length N: $p_1, \ldots, p_N$.

Among the integers not contained in the sequence $p_1, \ldots, p_N$ (not necessarily positive), find the integer nearest to X, that is, find the integer whose absolute difference with X is the minimum. If there are multiple such integers, report the smallest such integer.

Constraints

- $1 \leq X \leq 100$
- $0 \leq N \leq 100$
- $1 \leq p_i \leq 100$
- $p_1, \ldots, p_N$ are all distinct.
- All values in input are integers.

Input

Input is given from Standard Input in the following format:

X N
p_1 $\ldots$ p_N

Output

Print the answer.

Sample 1

Input	Output
6 5 4 7 10 6 5	8

Among the integers not contained in the sequence $4, 7, 10, 6, 5$, the one nearest to 6 is 8.

Sample 2

Input	Output
10 5 4 7 10 6 5	9

Among the integers not contained in the sequence $4, 7, 10, 6, 5$, the ones nearest to 10 are 9 and 11. We should print the smaller one, 9.

Sample 3

Input	Output
100 0	100

When $N = 0$, the second line in the input will be empty. Also, as seen here, X itself can be the answer.

Problem C. Bombs

Time Limit 2000 ms
Mem Limit 1048576 kB

Problem Statement

We have a board with R rows running horizontally and C columns running vertically. Let (i, j) denote the square at the i-th row from the top and j-th column from the left.

You are given characters $B_{i,j}$ representing the current states of (i, j). `.` represents an empty square; `#` represents a square with a wall; `1`, `2`,..., `9` represent a square with a bomb of power $1, 2, \ldots, 9$, respectively.

At the next moment, all bombs will explode **simultaneously**. When a bomb explodes, every square whose Manhattan distance from the square with the bomb is not greater than the power of the bomb will turn into an empty square. Here, the Manhattan distance from (r_1, c_1) to (r_2, c_2) is $|r_1 - r_2| + |c_1 - c_2|$.

Print the board after the explosions.

Constraints

- $1 \leq R, C \leq 20$
- R and C are integers.
- Each $B_{i,j}$ is one of `.`, `#`, `1`, `2`, ..., `9`.

Input

The input is given from Standard Input in the following format:

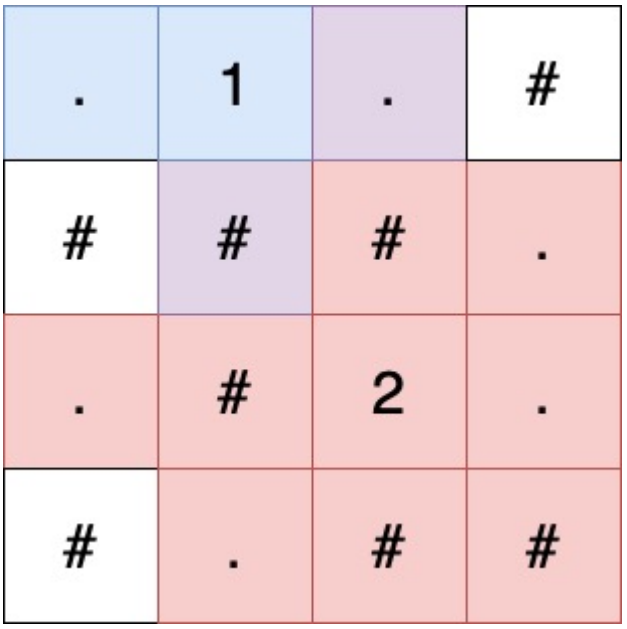
$$
\begin{array}{l}
R \ C \\
B_{1,1}B_{1,2}\ldots B_{1,C} \\
\vdots \\
B_{R,1}B_{R,2}\ldots B_{R,C}
\end{array}
$$

Output

Print R lines representing the board after the explosions. Use the format used in the input (do not print R or C).

Sample 1

Input	Output
4 4 .1.# ###. .#2. #.##	...# #... .... #...



- The explosion of the bomb at $(1, 2)$ will turn the blue squares and purple squares in the above figure into empty squares.
- The explosion of the bomb at $(3, 3)$ will turn the red squares and purple squares in the above figure into empty squares.

As seen in this sample, the explosion ranges of bombs may overlap.

Sample 2

Input	Output
2 5 ..#.# ###.#	..#.# ###.#

There may be no bombs.

Sample 3

Input	Output
2 3 11# ###	... ..#

Sample 4

Input	Output
4 6 #.#3#. ###.#. ##.### #1..#.	..... #..... #....# ....#.

Problem D. Good Distance

Time Limit 2000 ms
Mem Limit 1048576 kB

Problem Statement

There are N points in a D-dimensional space.

The coordinates of the i-th point are $(X_{i1}, X_{i2}, ..., X_{iD})$.

The distance between two points with coordinates $(y_1, y_2, ..., y_D)$ and $(z_1, z_2, ..., z_D)$ is $\sqrt{(y_1 - z_1)^2 + (y_2 - z_2)^2 + ... + (y_D - z_D)^2}$.

How many pairs (i, j) $(i < j)$ are there such that the distance between the i-th point and the j-th point is an integer?

Constraints

- All values in input are integers.
- $2 \leq N \leq 10$
- $1 \leq D \leq 10$
- $-20 \leq X_{ij} \leq 20$
- No two given points have the same coordinates. That is, if $i \neq j$, there exists k such that $X_{ik} \neq X_{jk}$.

Input

Input is given from Standard Input in the following format:

$$
\begin{array}{l}
N \ \ D \\
X_{11} \ \ X_{12} \ \ ... \ \ X_{1D} \\
X_{21} \ \ X_{22} \ \ ... \ \ X_{2D} \\
\vdots \\
X_{N1} \ \ X_{N2} \ \ ... \ \ X_{ND}
\end{array}
$$

Output

Print the number of pairs (i, j) $(i < j)$ such that the distance between the i-th point and the j-th point is an integer.

Sample 1

Input	Output
3 2 1 2 5 5 -2 8	1

The number of pairs with an integer distance is one, as follows:

- The distance between the first point and the second point is $\sqrt{|1-5|^2+|2-5|^2} = 5$, which is an integer.
- The distance between the second point and the third point is $\sqrt{|5-(-2)|^2+|5-8|^2} = \sqrt{58}$, which is not an integer.
- The distance between the third point and the first point is $\sqrt{|-2-1|^2+|8-2|^2} = 3\sqrt{5}$, which is not an integer.

Sample 2

Input	Output
3 4 -3 7 8 2 -12 1 10 2 -2 8 9 3	2

Sample 3

Input	Output
5 1 1 2 3 4 5	10

Problem E. Substring

Time Limit 2000 ms
Mem Limit 1048576 kB

Problem Statement

Given are two strings S and T.

Let us change some of the characters in S so that T will be a substring of S.

At least how many characters do we need to change?

Here, a substring is a consecutive subsequence. For example, `xxx` is a substring of `yxxxy`, but not a substring of `xxyxx`.

Constraints

- The lengths of S and T are each at least 1 and at most 1000.
- The length of T is at most that of S.
- S and T consist of lowercase English letters.

Input

Input is given from Standard Input in the following format:

S
T

Output

Print the minimum number of characters in S that need to be changed.

Sample 1

Input	Output
cabacc abc	1

For example, changing the fourth character `a` in S to `c` will match the second through fourth characters in S to T.

Since S itself does not have T as its substring, this number of changes - one - is the minimum needed.

Sample 2

Input	Output
`codeforces` `atcoder`	`6`

Problem F. Knight Fork

Time Limit 2000 ms
Mem Limit 1048576 kB

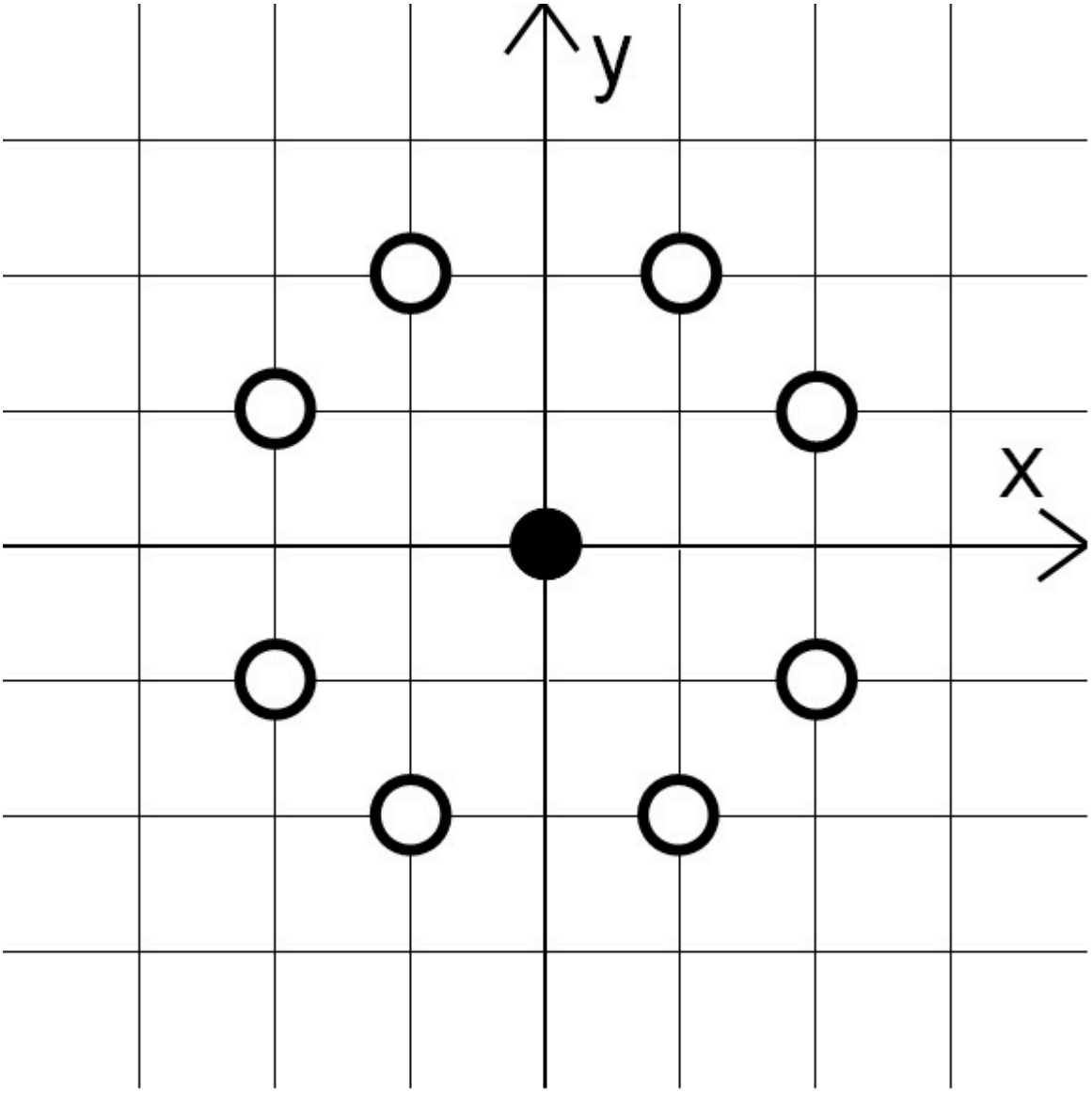
Problem Statement

On an xy-coordinate plane, is there a lattice point whose distances from two lattice points (x_1, y_1) and (x_2, y_2) are both $\sqrt{5}$?

Notes

A point on an xy-coordinate plane whose x and y coordinates are both integers is called a lattice point.
The distance between two points (a, b) and (c, d) is defined to be the Euclidean distance between them, $\sqrt{(a-c)^2 + (b-d)^2}$.

The following figure illustrates an xy-plane with a black circle at $(0, 0)$ and white circles at the lattice points whose distances from $(0, 0)$ are $\sqrt{5}$. (The grid shows where either x or y is an integer.)



Constraints

- $-10^9 \leq x_1 \leq 10^9$
- $-10^9 \leq y_1 \leq 10^9$
- $-10^9 \leq x_2 \leq 10^9$
- $-10^9 \leq y_2 \leq 10^9$
- $(x_1, y_1) \neq (x_2, y_2)$
- All values in input are integers.

Input

Input is given from Standard Input in the following format:

x_1 y_1 x_2 y_2

Output

If there is a lattice point satisfying the condition, print `Yes`; otherwise, print `No`.

Sample 1

Input	Output
`0 0 3 3`	`Yes`

- The distance between points $(2, 1)$ and (x_1, y_1) is $\sqrt{(0-2)^2+(0-1)^2} = \sqrt{5}$;
- the distance between points $(2, 1)$ and (x_2, y_2) is $\sqrt{(3-2)^2+(3-1)^2} = \sqrt{5}$;
- point $(2, 1)$ is a lattice point,

so point $(2, 1)$ satisfies the condition. Thus, `Yes` should be printed.
One can also assert in the same way that $(1, 2)$ also satisfies the condition.

Sample 2

Input	Output
`0 1 2 3`	`No`

No lattice point satisfies the condition, so `No` should be printed.

Sample 3

Input	Output
`1000000000 1000000000 999999999 999999999`	`Yes`

Point $(10^9+1, 10^9-2)$ and point $(10^9-2, 10^9+1)$ satisfy the condition.

Problem G. TaK Code

Time Limit 2000 ms
Mem Limit 1048576 kB

Problem Statement

Takahashi invented Tak Code, a two-dimensional code. A TaK Code satisfies all of the following conditions:

- It is a region consisting of nine horizontal rows and nine vertical columns.
- All the 18 cells in the top-left and bottom-right three-by-three regions are black.
- All the 14 cells that are adjacent (horizontally, vertically, or diagonally) to the top-left or bottom-right three-by-three region are white.

It is not allowed to rotate a TaK Code.

You are given a grid with N horizontal rows and M vertical columns. The state of the grid is described by N strings, $S_1, \ldots,$ and S_N, each of length M. The cell at the i-th row from the top and j-th column from the left is black if the j-th character of S_i is `#`, and white if it is `.`.

Find all the nine-by-nine regions, completely contained in the grid, that satisfy the conditions of a TaK Code.

Constraints

- $9 \leq N, M \leq 100$
- N and M are integers.
- S_i is a string of length M consisting of `.` and `#`.

Input

The input is given from Standard Input in the following format:

```
N M
S_1
```

$\vdots$
S_N

Output

For all pairs (i, j) such that the nine-by-nine region, whose top-left cell is at the i-th row from the top and j-th columns from the left, satisfies the conditions of a TaK Code, print a line containing i, a space, and j in this order.
The pairs must be sorted in lexicographical ascending order; that is, i must be in ascending order, and within the same i, j must be in ascending order.

Sample 1

Input	Output
19 18 ###.....###..... ###.....###..... ###..#...###..#... .....#... ....###.....### .....###.....### .....###.....### .###..... .###.....##..... .###..... .....###... ...##.....###... ...##.....###... .....###..... .....###..... .....###..... .....#.....	1 1 1 10 7 7 10 2

A TaK Code looks like the following, where `#` is a black cell, `.` is a white cell, and `?` can be either black or white.

```
###.?????
###.?????
###.?????
....?????
?????????
?????....
?????.###
?????.###
?????.###
```

In the grid given by the input, the nine-by-nine region, whose top-left cell is at the 10-th row from the top and 2-nd column from the left, satisfies the conditions of a TaK Code, as shown below.

```
###......
###......
###......
.........
..##.....
..##.....
......###
......###
......###
```

Sample 2

Input	Output
9 21 ###.#.....#.### ###.#.....#.### ###.#.....#.### ....#.....#.... #########...######### ....#.....#.... ....#.###...###.#.... ....#.###...###.#.... ....#.###...###.#....	1 1

Sample 3

Input	Output
18 18 ######..... ######..... ######..... ######..... ######..... ######..... ....###### .....###### .....###### .....###### .....###### .....######	

There may be no region that satisfies the conditions of TaK Code.

Problem H. Exponential

Time Limit 2000 ms
Mem Limit 1048576 kB

Problem Statement

You are given a positive integer X. Find the largest *perfect power* that is at most X. Here, a perfect power is an integer that can be represented as b^p, where b is an integer not less than 1 and p is an integer not less than 2.

Constraints

- $1 \leq X \leq 1000$
- X is an integer.

Input

Input is given from Standard Input in the following format:

```
X
```

Output

Print the largest perfect power that is at most X.

Sample 1

Input	Output
10	9

There are four perfect powers that are at most 10: 1, 4, 8 and 9. We should print the largest among them, 9.

Sample 2

Input	Output
1	1

Sample 3

Input	Output
999	961

Problem I. Snack

Time Limit 2000 ms
Mem Limit 1048576 kB

Problem Statement

Takahashi is organizing a party.

At the party, each guest will receive one or more snack pieces.

Takahashi predicts that the number of guests at this party will be A or B.

Find the minimum number of pieces that can be evenly distributed to the guests in both of the cases predicted.

We assume that a piece cannot be divided and distributed to multiple guests.

Constraints

- $1 \leq A, B \leq 10^5$
- $A \neq B$
- All values in input are integers.

Input

Input is given from Standard Input in the following format:

```
A B
```

Output

Print the minimum number of pieces that can be evenly distributed to the guests in both of the cases with A guests and B guests.

Sample 1

Input	Output
2 3	6

When we have six snack pieces, each guest can take three pieces if we have two guests, and each guest can take two if we have three guests.

Sample 2

Input	Output
123 456	18696

Sample 3

Input	Output
100000 99999	9999900000

Problem J. Triangle?

Time Limit 2000 ms
Mem Limit 1048576 kB

Problem Statement

In the xy-plane, we have N points numbered 1 through N.
Point i is at the coordinates (X_i, Y_i). Any two different points are at different positions.
Find the number of ways to choose three of these N points so that connecting the chosen points with segments results in a triangle with a positive area.

Constraints

- All values in input are integers.
- $3 \leq N \leq 300$
- $-10^9 \leq X_i, Y_i \leq 10^9$
- $(X_i, Y_i) \neq (X_j, Y_j)$ if $i \neq j$.

Input

Input is given from Standard Input in the following format:

$$
\begin{array}{l}
N \\
X_1 \ \ Y_1 \\
X_2 \ \ Y_2 \\
\ldots \\
X_N \ \ Y_N
\end{array}
$$

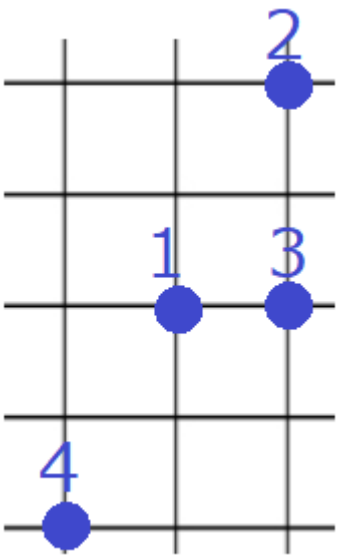
Output

Print the answer as an integer.

Sample 1

Input	Output
4 0 1 1 3 1 1 -1 -1	3

The figure below illustrates the points.



There are three ways to choose points that form a triangle: $\{1, 2, 3\}, \{1, 3, 4\}, \{2, 3, 4\}$.

Sample 2

Input	Output
20 224 433 987654321 987654321 2 0 6 4 314159265 358979323 0 0 -123456789 123456789 -1000000000 1000000000 124 233 9 -6 -4 0 9 5 -7 3 333333333 -333333333 -9 -1 7 -10 -1 5 324 633 1000000000 -1000000000 20 0	1124